

The Long Road to Cooperstown: Exploring Baseball Hall of Fame Voting

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GitHub repository: https://github.com/rubash9849/Data_Science_Practice_CS5830

Presentation slides:

<https://docs.google.com/presentation/d/1HbDt0n2bOA5vvpPvIFtmSaF0hOwQvw0f/edit?usp=sharing&oid=104804874161619715250&rtpof=true&sd=true>

1. Introduction

The Baseball Hall of Fame recognizes players, managers, umpires, and other individuals who made important contributions to professional baseball. However, reaching the Hall of Fame is not a simple or immediate process. A candidate may appear in several election years, receive support from different voting organizations, and still never be inducted. For baseball historians, journalists, voters, and fans, understanding these different paths provides a clearer picture of how careers are remembered and recognized.

We explored Hall of Fame voting records to understand who conducted the voting, how often induction occurred, and how candidates' paths differed. We also examined salary data to see whether a player who was not inducted could still receive financial recognition. Our results show that Hall of Fame induction was uncommon and that there was no single path to baseball recognition.

2. Dataset

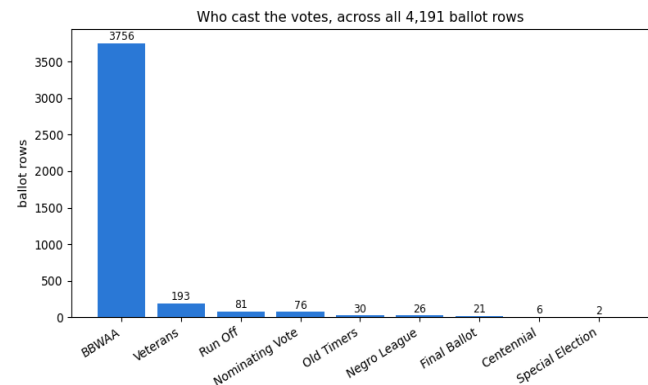
The primary dataset, HallOfFame.csv, contains 4,191 records and nine variables. Each row represents one candidate's Hall of Fame record in one election year. Therefore, the dataset does not contain 4,191 different people because the same person can appear in multiple years. Important variables include the candidate identification code, election year, voting organization, total ballots, votes required, votes received, induction result, and professional category. The inducted column uses Y for a successful induction and N for an unsuccessful record. The category field identifies whether the person was considered as a player, manager, umpire, or pioneer/executive. We connected this dataset with People.csv to obtain readable names and career dates. We also used Salaries.csv for a limited comparison between one never-inducted player's annual salary and the MLB mean. Together, these attributes allowed us to examine Hall of Fame selectivity, candidate voting histories, and an alternative form of career recognition.

3. Analysis Technique

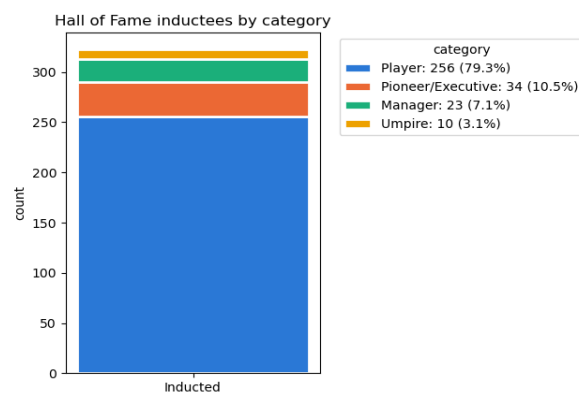
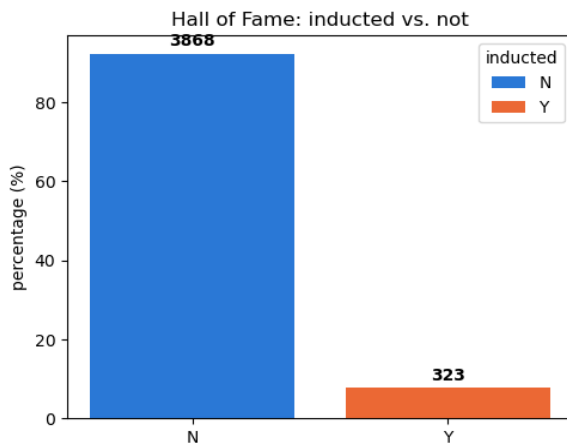
We used exploratory data analysis because our goal was to describe historical patterns rather than predict future induction results. Counts and percentages were used to compare voting organizations, induction outcomes, and professional categories. Candidates' records were grouped by identification code to examine how frequently each person appeared in the Hall of Fame data. Bar charts were appropriate for comparing categories and record counts, while line charts were used to show how vote percentages and salaries changed over time. These techniques were suitable because they presented the results clearly to readers without requiring advanced statistical knowledge. They also allowed us to move from broad patterns in the full dataset to individual stories representing successful, unsuccessful, and financially rewarded careers.

4. Results

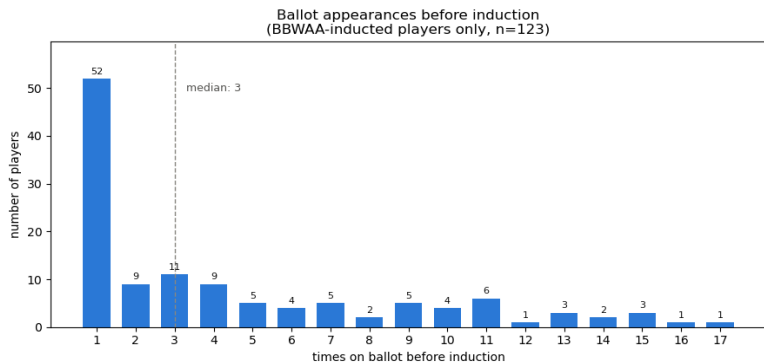
The BBWAA contributed 3,756 of the 4,191 records, or approximately 89.6%, making it the main voting organization represented in the dataset. This means that the overall findings mostly reflect voting by baseball writers, although other groups, such as the Veterans Committee, also played a role in selecting Hall of Fame members.



Induction was uncommon. Only 323 records, or 7.7%, resulted in induction, compared with 3,868 unsuccessful records. Among the successful records, players formed the largest group, accounting for 256 of the 323 inductions, or approximately 79.3%. This shows that simply appearing in a Hall of Fame record did not usually lead to induction. It also shows that the Hall mainly recognized players, while managers, umpires, and pioneers or executives represented smaller groups.



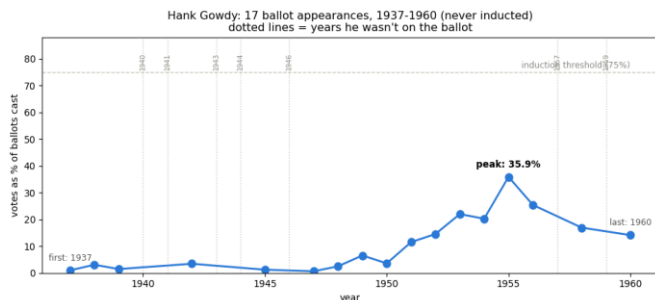
Successful candidates also followed different timelines. Among the 123 candidates inducted by the BBWAA in our analysis, 52 had one recorded appearance, making it the most common individual result. However, the median was three appearances, and a smaller number remained in the records for much longer.



Edd Roush represented the longest successful path, with 20 election-year records before eventual induction by the Veterans Committee in 1962. His case shows that recognition could come much later and through a different voting organization.

playerID	player	times_on_ballot
roushed01	Edd Roush	20
youngro01	Ross Youngs	19
schalra01	Ray Schalk	18
ruffire01	Red Ruffing	18
wheatza01	Zack Wheat	17
rizzuph01	Phil Rizzuto	17
wilsoha01	Hack Wilson	17
rixeyep01	Eppa Rixey	17
faberre01	Red Faber	17
vanceda01	Dazzy Vance	17
reesepe01	Pee Wee Reese	17
lazzeto01	Tony Lazzeri	16
bendeck01	Chief Bender	16
slaugen01	Enos Slaughter	16
santoro01	Ron Santo	16

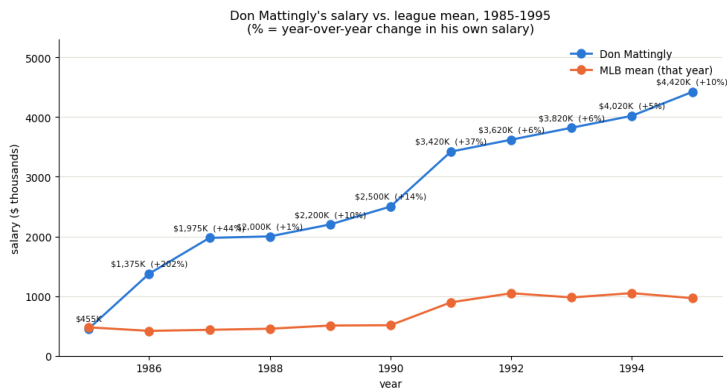
A long voting history did not always end successfully. Hank Gowdy had 17 election-year records, the highest count among the never-inducted candidates examined, but his highest displayed support was only 35.9%. This was below the approximately 75% threshold shown in the chart. When compared with Edd Roush, Gowdy's history shows that the number of appearances alone did not reveal how close a candidate was to induction. Voting support and the organization conducting the election also mattered.



Finally, we checked salary availability for 15 frequently considered candidates who were never inducted. Only four had salary records because the salary dataset begins in 1985. Don Mattingly had the most complete information, with 11 recorded seasons from 1985 through 1995. During this period, his salary increased from approximately **\$455,000 to \$4.42 million** and was above the MLB mean during much of the period. Although salary does not directly measure talent, this

playerID	debut	finalGame	player	salary_rows
mattido01	1982-09-08	1995-10-01	Don Mattingly	11
murphda05	1976-09-13	1993-05-21	Dale Murphy	9
johnho01	1963-09-06	1989-05-25	Tommy John	5
garvest01	1969-09-01	1987-05-23	Steve Garvey	3
gowdyha01	1910-09-13	1930-08-29	Hank Gowdy	0
vernomo01	1939-07-08	1960-09-27	Mickey Vernon	0
newcodo01	1949-05-20	1960-10-01	Don Newcombe	0
darkal01	1946-07-14	1960-10-02	Al Dark	0
marisro01	1957-04-16	1968-09-29	Roger Maris	0
kluszte01	1947-04-18	1961-10-01	Ted Kluszewski	0
thomsbo01	1946-09-09	1960-07-17	Bobby Thomson	0
tiantlu01	1964-07-19	1982-09-04	Luis Tiant	0
munsoth01	1969-08-08	1979-08-01	Thurman Munson	0
floodcu01	1956-09-09	1971-04-25	Curt Flood	0
willsmo01	1959-06-06	1972-10-04	Maury Wills	0

example shows that a player could receive substantial financial recognition without Hall of Fame induction.



Overall, the records show that there was no single path to baseball recognition. Some candidates were inducted quickly, some waited many years, and others were never inducted despite repeated consideration. Hall of Fame induction was an important honor, but it was not the only indication that a baseball career was valued.

5. Technical

The analysis was conducted in Python using pandas for loading, inspecting, filtering, combining, and summarizing the datasets. Matplotlib was used to create the bar and line charts. We first examined the dimensions, column names, data types, missing values, and sample rows in `HallOfFame.csv`. Candidate identification, election year, voting organization, induction status, and professional category were available for all relevant records. Some values were missing from the ballots, required-votes, and received-votes fields because numerical voting totals were not reported for every election process. Records with missing numerical values were retained for categorical counts but were not used when a vote percentage required those values. We joined `HallOfFame.csv` with `People.csv` through the shared `playerID` field to display names and career information.

The analyses used `value_counts()` to summarize categories, Boolean filters to select successful and unsuccessful records, and `groupby()` to count records associated with each candidate. Bar charts were used for voting organizations, induction outcomes, categories, and appearance distributions. Line charts were used for voting percentages and salary changes because the order of years was important. One limitation is that the salary dataset covers only 1985 through 2016, leaving no salary information for most older candidates. Don Mattingly was chosen because he had the best available salary coverage among the selected never-inducted candidates. Future work could separate voting organizations more strictly, compare maximum vote percentages, and adjust historical salaries for inflation.